TidyTuesday Week 45

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Monday 2nd November, 2026

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1 Weekly Summary

AI-Generated Content

The summary below was written by Claude (Anthropic) from that week's regression outputs and series descriptions. It has not been reviewed or fact-checked by a human, and should be read as an automated, informal recap – not analysis.

This week was a quiet one, by which I mean the random number generator reached into the vast bucket of the entire American economy and produced exactly one pairing worth agonizing over. On the menu: the number of goods-producing employees in Springfield, Massachusetts, versus a price index for proprietary and government nursing homes. Because of course those two things belong in a sentence together. If you've ever lain awake wondering whether factory headcounts in western Massachusetts secretly dance with the going rate of nursing home care nationwide, well, strap in.

The raw regression came out absolutely gleaming. An R-squared of 0.78, a p-value with enough zeros after the decimal point to make a scientist spill their coffee, and a confidently negative slope suggesting that as Springfield's goods-producing jobs went down, nursing home prices went up. Dramatic! Compelling! Nobel-adjacent! I was already drafting the press release about the Springfield Manufacturing Theory of Elder Care Inflation.

And then, as it always does, the detrending stepped in to ruin the party. Once we stripped out each series' own long-run drift – Springfield's factory jobs slowly shrinking over decades, nursing home prices slowly climbing over decades, as prices and employment trends are wont to do independently of one another – the glorious relationship evaporated into nothing. The R-squared cratered to a frankly insulting 0.02, and the p-value strolled up to 0.52, which in statistics is roughly the equivalent of a shrug. The whole "link" was just two lines drifting in opposite directions over time and the math mistaking that for romance. Textbook trend mirage.

As always, the reminder: these pairs are picked at random, there is no theory here, no causal claim, and no reason on Earth that Springfield's job market should know or care about nursing home billing. A collapse like this week's is not the exception around here – it's the entire point. Spurious correlation is the house specialty, and this week the kitchen delivered a perfect example. See you next week for more statistically significant nonsense.

2 Date: 2026-11-02

Series ID: SMU25781000600000001
This series is titled All Employees: Goods Producing in Springfield, MA-CT (NECTA) and has a frequency of Monthly. The units are Thousands of Persons and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 1990-01-01 and the observation end date is 2024-12-01.The popularity of this series is 1.

Series ID: PRGONUPIHCSA
This series is titled Medical Services Expenditures by Provider: Nursing Homes: Proprietary and Government Nursing Homes Price Index and has a frequency of Annual. The units are Index 2017=100 and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 2000-01-01 and the observation end date is 2021-01-01.The popularity of this series is 1.

2.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_PRGONUPIHCSA	**R-squared:**	0.783
Model:	OLS	**Adj. R-squared:**	0.772
Method:	Least Squares	**F-statistic:**	72.01
Date:	Mon, 02 Nov 2026	**Prob (F-statistic):**	4.64e-08
Time:	21:06:51	**Log-Likelihood:**	-73.979
No. Observations:	22	**AIC:**	152.0
Df Residuals:	20	**BIC:**	154.1
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	186.2043	11.787	15.797	0.000	161.616	210.793
value_fred_SMU25781000600000001	-2.2652	0.267	-8.486	0.000	-2.822	-1.708

Omnibus:	5.822	**Durbin-Watson:**	0.251
Prob(Omnibus):	0.054	**Jarque-Bera (JB):**	4.103
Skew:	1.043	**Prob(JB):**	0.129
Kurtosis:	3.353	**Cond. No.**	333.

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

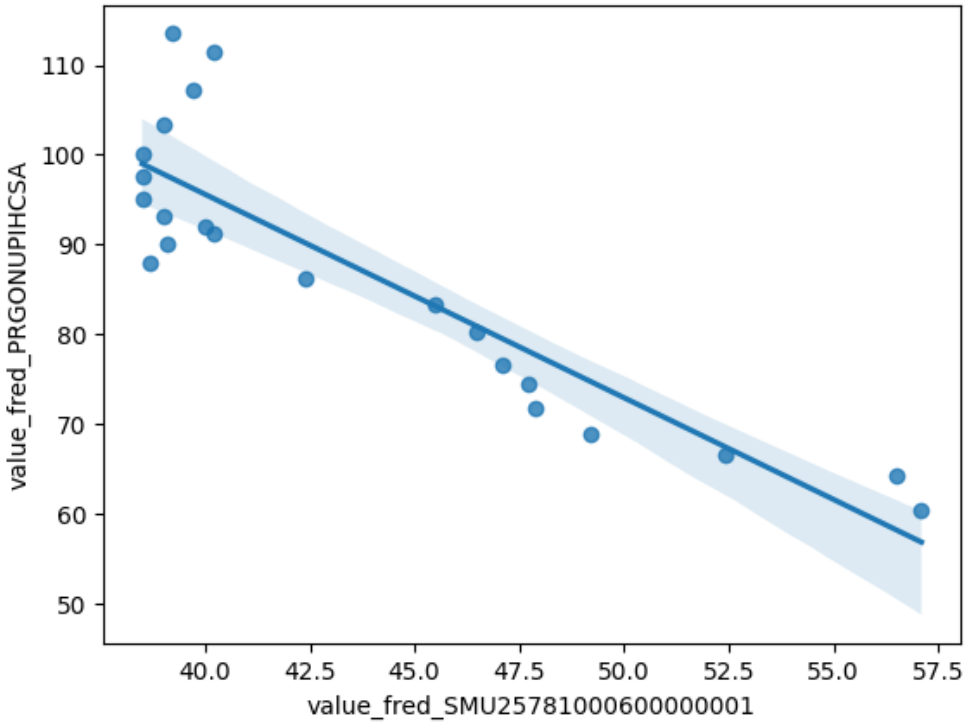


Figure 1: Raw Regression Plot for 2026-11-02

2.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_PRGONUPIHCSA_detrended	**R-squared:**	0.021
Model:	OLS	**Adj. R-squared:**	-0.028
Method:	Least Squares	**F-statistic:**	0.4223
Date:	Mon, 02 Nov 2026	**Prob (F-statistic):**	0.523
Time:	21:06:51	**Log-Likelihood:**	-42.096
No. Observations:	22	**AIC:**	88.19
Df Residuals:	20	**BIC:**	90.37
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	4.027e-13	0.367	1.1e-12	1.000	-0.765	0.765
value_fred_SMU25781000600000001_detrended	-0.0866	0.133	-0.650	0.523	-0.365	0.191

Omnibus:	2.262	**Durbin-Watson:**	0.331
Prob(Omnibus):	0.323	**Jarque-Bera (JB):**	1.135
Skew:	-0.097	**Prob(JB):**	0.567
Kurtosis:	1.904	**Cond. No.**	2.75

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

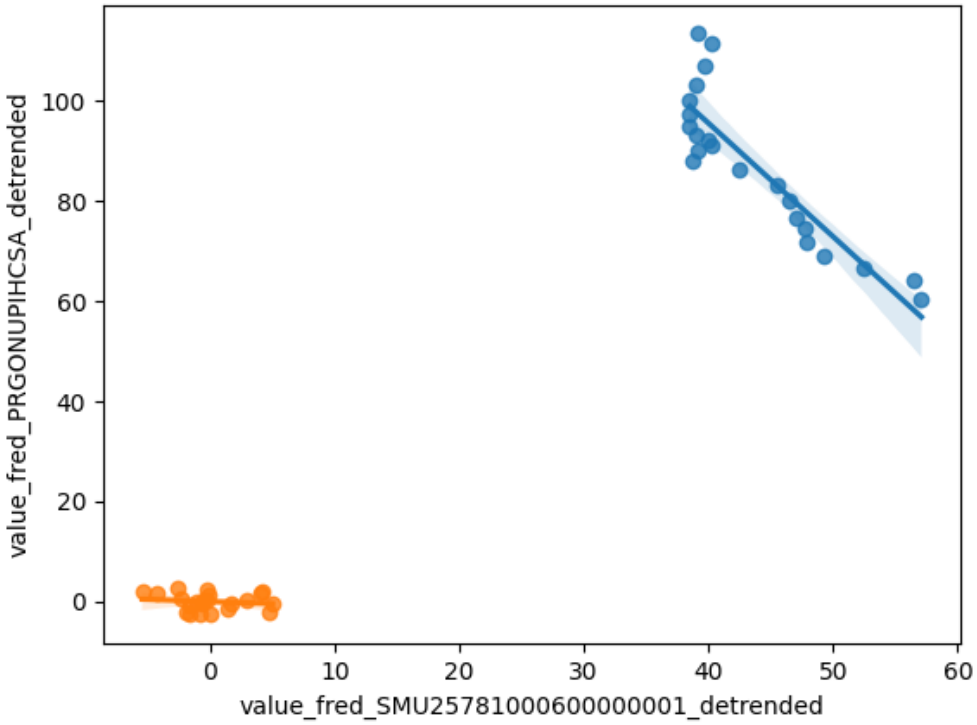


Figure 2: Detrended Regression Plot for 2026-11-02